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MILITARY SPECIFICATION

ALTIMETER SET, ELECTRONIC, AN/APN-171(V)

This specification is approved by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, and acceptance requirements for the AN/APN 171(V) electronic altimeter set, herein referred to as the equipment. The equipment covered by this specification shall provide an accurate indication of absolute altitude of an aircraft over land and water.

*1.2 Classification. The equipment covered by this specification consists of the following items:

Item	Type Designation
Receiver-Transmitter, Radar	RT-804B/APN-171(V)
Receiver-Transmitter, Radar	RT-805/APN-171(V)
Receiver-Transmitter, Radar	RT-806A/APN-171(V)
Receiver-Transmitter, Radar	RT-829A/APN-171(V)
Receiver-Transmitter, Radar	RT-1006/APN-171(V)
Receiver-Transmitter, Radar	RT-1062A/APN-171(V)
Indicator, Height	ID-1345/APN-171(V)
Indicator, Height	ID-1346/APN-171(V)
Indicator, Height	ID-1725/APN-171(V)
Antenna	AS-1858/APN-171(V)
Mounting	MT-4167/APN-171(V)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5826

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1.3 Associated Equipment. The equipment should operate with the associated equipment listed in 6.8.

2. APPLICABLE DOCUMENTS

*2.1 Government Documents.

2.1.1 Specifications and Standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-675	Coating of Glass Optical Elements (Anti Reflection)
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference control Requirements, Aircraft Equipments
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Acceptance, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-L-27160	Lighting, Instrument, Integral White, General Specification for

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STANDARDS

MILITARY

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts & Equipment, Procedures for Packaging and Packing of
MIL-STD-965	Parts Control Program
MIL-STD-2077	Test Program, Set General Requirement for
MIL-STD-2074	Failure Classification for Reliability Testing
MS 17322	Meter, Time Totalizing, 115 Volt, 400 cycle, Rev. B
MS 24367	Lamps, Incandescent, Miniature Integral Lighting

2.1.2 Other Government documents, drawings, and publications. The following publications form a part of this specification to the extent specified herein.

PUBLICATIONS

Naval Air Systems Command

EI-562	Avionics, Installation Instructions for Altimeter Set, Electronic, AN/APN-171(V) and Altimeter Set, Electronic, AN/APN-171B(V)
ET-562	Avionics Bench, Preflight, and Flight Test Instructions for Altimeter Set, Electronic, AN/APN-171(V) and Altimeter Set, Electronic, AN/APN-171B(V)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 First article. When specified, a sample(s) shall be subjected to First Article Inspection (See 4.2).

3.2 Parts and materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Microelectronic Technology shall be considered and microelectronic items shall conform to requirements specified herein.

b. Other parts and materials requirements shall conform to MIL-E-5400.

c. Nonrepairable subassemblies, as specified in MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies shall be reliable. (See 6.5).

d. When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard parts and materials approval. Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diode) other than microelectronic devices shall be obtained as specified in MIL-E-5400. Microelectronic devices shall be approved as specified in MIL-STD-965.

3.2.2 Microelectronic modular assemblies. When used, Microelectronic Modular Assemblies shall meet the requirements of MIL-STD-965.

3.3 Design and construction. The equipment shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

*3.3.1 Total weight. The total weight of the system, excluding cables, shall be not greater than 20.8 pounds. (System weight is based on the use of a receiver-transmitter, mounting rack, two height indicators and two antennas.)

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3.3.2 Reliability

3.3.2.1 Operational stability. The equipment shall operate with prescribed performance, continuously or intermittently for a period of at least 500 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating life. The equipment shall have a total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in mean time between failure (MTBF). The equipment shall have 500 hours of mean (operating) time between failures when tested and accepted as specified in 4.4.3.

3.3.2.4 Time totalizing meter. The Receiver-Transmitters shall contain time totalizing meters in accordance with MIL-M-7793 and MS 17322.

3.3.3 Cabling and connections

3.3.3.1 Cables and connectors. The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.3.3.2 Interconnection cabling. The equipment shall be capable of prescribed operation using external wiring in accordance with MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Control panels. No control panels or pilot controls, other than those integral with the indicator, shall be required.

3.3.5 Interchangeability. The equipment shall meet the interchangeability requirements of MIL-E-5400.

3.3.6 Interference control. The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-I-6181 except as modified below in 3.3.6.1 and 3.3.6.2.

3.3.6.1 Receiver-Port radiation. Since the local oscillator frequency is the same as the transmitter frequency, compliance with the local oscillator radiation requirements of MIL-I-6181 shall not be required for the frequency band between 4290 and 4310 megahertz.

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3.3.6.2 Transmitter key down spurious emanations. The transmitter key down spurious emanations, (see of MIL-I-6181) shall be measured as radiated from the equipment antenna and shall be within the limits shown on Figure 1. The peak power output of the second harmonic shall be not greater than 1 watt.

3.3.7 Maintenance provisions and field testing. Provisions for maintenance shall be as prescribed in MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.7.1 VAST test program sets. When specified in the contract, Versatile Avionic Shop Test System (VAST) Test Program Sets shall be furnished in accordance with MIL-STD-2077. If VAST Test Program Sets exist for the equipment, and changes to the equipment are made which will affect the fault diagnosis procedure, changes to the existing Test Program Sets shall be prepared as part of the equipment changes in accordance with MIL-STD-2077.

3.3.8 Nomenclature and nameplates. Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.3.9 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

a.	Temperature	Room ambient ($25^{o}C \pm 5^{o}C$)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90% relative humidity
e.	Input Power Voltage	115 ± 1.0 VAC 27.5 ± 0.5 VDC

3.3.10 Service conditions. The equipment shall operate with prescribed performance under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

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3.3.10.1 Vibration

3.3.10.1.1 Receiver-Transmitter. The receiver-transmitter, when mounted on a MT-4167/APN-171 mounting rack, shall operate with prescribed performance when subjected to the vibration requirements of Curve IV of MIL-E-5400. The receiver-transmitter, when hard mounted, shall operate with prescribed performance when subjected to the vibration requirements of curve II of MIL-E-5400.

3.3.10.1.2 Indicator. The indicator, when hard-mounted, shall operate with prescribed performance when subjected to the vibration requirements of curve III of MIL-E-5400.

3.3.10.1.3 Antenna. The antenna shall operate with prescribed performance when subjected to the vibration requirements of Curve IV of MIL-E-5400.

3.3.10.2 Temperature. Operational temperature for the equipment shall be not greater than +71°C.

3.3.10.3 Altitude. Operational altitude for the equipment shall be not greater than 50,000 feet.

3.3.11 Warm-up time. The time required for the equipment to warm-up prior to operation shall be not greater than 3 minutes under standard conditions and extreme service conditions.

3.3.12 Input electrical power

3.3.12.1 Operating power. The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall be not greater than the specified amounts.

a. AC Power (single Phase), 115V, Category B, 100VA

b. DC Power 28V Category B, 0.25 Watt per Indicator

3.3.12.2 Lighting power. Input power for lighting shall be not greater than 0.5 amp at 5 volts, or 0.35 ampere at 28 volts.

3.3.12.3 Degraded performance. Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

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3.4 Performance. Unless otherwise specified herein, values set forth to establish the requirements of prescribed performance apply to performance under standard, extreme service, and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation. The altimeter set shall be a range tracking device that utilizes pulse techniques and leading edge tracking of ground return signals to produce accurate altitude information. The set has search and track modes of operation. In the search mode, the set successively examines the altitude range at a search rate of approximately three complete cycles per second. In the track mode, the system locks on and tracks a ground return and gives continuous altitude information. Altitude information is obtained by measuring the time from the instant the RF energy leaves the transmitter until the first ground return signal is detected in the receiver. The system converts this time interval information to a linear dc voltage. The linear dc voltage is supplied to operate the indicator and is applied to various circuits within the receiver-transmitter to provide other types of output signals.

3.4.2 Altitude range. The set shall indicate absolute altitude from 0 to 5000 feet or 0 to 1000 feet over water or land.

3.4.2.1 Altitude accuracy

★3.4.2.1.1 Standard conditions. The receiver-transmitter altitude output signals of 3.5.1.7.2.1 herein shall be within 3 ft. plus 1 percent of the actual altitude below 200 ft., and 1.5 ft. plus 1 percent above 200 ft.

3.4.2.1.2 Service conditions. The receiver-transmitter output altitude signals of 3.5.1.7.2.1 herein shall be within 5 ft. plus 3 percent of the actual altitude plus 5 percent of the average range rate. The indicator shall be capable of displaying altitude in feet within the accuracy requirements of 3.5.5.1 and 3.5.6.1 herein.

3.4.2.2 Altitude resolution. For changes in altitude as measured over a flat surface, the minimum altitude resolution of the receiver-transmitter shall be 6 inches or 0.1 percent of actual altitude, whichever is greater. The indicator resolution shall be within the sensitivity requirements of 3.5.5.1 and 3.5.6.1 herein.

3.4.2.3 Altitude memory. The "in-track" output of the receiver-transmitter described in s 3.5.1.7.2.1 herein shall be maintained for 1 ± 0.5 seconds after loss of return signal before reverting to the "out of track" state.

3.4.3 Frequency. The system shall operate at a center frequency of 4300 ± 10 megahertz.

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3.4.4 Range rate tracking capability. The system shall be capable of maintaining continuous track for changes in range occurring at rates of greater than ± 2,000 feet/second or less.

3.4.5 Ground speed. The system shall be capable of specified operation and meeting the accuracy requirements while operating over terrain or water at any ground speed including Mach 2.0.

*3.4.6 Sensitivity. The equipment shall be capable of meeting the requirements of this specification with a signal attenuation not greater than 121 dB at 5,000 feet of altitude, 115 dB at 1000 feet.

3.4.7 Aircraft attitude. The set shall meet the requirements of this specification in an aircraft when climbing and diving at 30° angles and when banking at 30° angles for an encountered coefficient of reflection of 0.08 and greater. No detrimental effects shall be caused to the operation of the set by any flight maneuver, and operation of the set in any position shall not prevent normal operation when the aircraft returns to normal flight.

3.4.8 False indication. The system shall not lock-on any interferring non-synchronous false signals or noise.

3.4.9 Low altitude limit function. Low altitude limit functions shall be as follows:

a. A switching function shall be provided in the indicator to provide a warning when the aircraft altitude becomes less than a pre-selected height. The switching function shall close and open within the tolerance requirements of 3.5.5.1 and 3.5.6.1 herein. The low altitude warning shall be given by a warning light on the indicator as shown in Figures 2 and 3. The legend shall be black on a yellow background with a minimum of two lamps used to illuminate the legend. When not energized, the legend shall not be readable and shall not appear energized (e.g., in bright sunlight). The low altitude limit index shall be adjustable from 0 to maximum altitude.

b. The ground lead of the low altitude light shall not be grounded internally but shall be brought out to the electrical connector for the purpose of externally dimming the light.

c. The low altitude warning circuit shall not operate when (1) the signal strength becomes inadequate, or (2) the system is turned off, or (3) the altitude is above the maximum altitude range. The circuit shall also be capable of providing 0.42 amp 28 VDC for external use.

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*3.4.10 Aircraft installation delay adjustment. The Receiver-Transmitter shall be adjustable to allow for installation of the antenna in the aircraft from 1 to 10 feet above the ground and total RF cable lengths of 5 to 20 feet of RG214/U or equivalent flexible coaxial cable or 10 to 60 feet of RG231/U or equivalent semi-rigid coaxial cable. This adjustment is not externally accessible.

3.4.11 Self-Test. Provisions shall be incorporated to permit go-no-go confidence testing of the electronic altimeter set so that rapid ground or air check of the equipment can be accomplished. The confidence check is initiated by pressing the control knob on the indicator. Satisfactory system operation is indicated by correct readout of the simulated altitude plus correct readout of zero $\pm$ 5 feet when the test button is released on the ground. An indicator readout of 100 $\pm$ 15 feet shall result during the period of the confidence test. Self test shall be capable of being energized continuously with no detrimental effect to the equipment. The self test shall be mechanized by injecting a delayed detected transmitter pulse to simulate normal operation of the system. Under standard conditions, the repeatability of the test altitude indication shall be within 3 feet for each installation.

3.5 Detail requirements

3.5.1 Receiver-Transmitters. Each Receiver-Transmitter unit shall meet all applicable system requirements and the detail requirements specified in 3.5.1.1 through 3.5.12.7.

3.5.1.1 Function. The Receiver-Transmitter shall transmit a radio frequency signal, receive the reflected signal, and derive the proper analog voltage for 0 to 5,000 feet or 0 to 1,000 feet of radar altitude in accordance with the requirements of this specification.

*3.5.1.2 Form factor. The overall maximum dimensions for the Receiver-Transmitter shall be contained within the volume of EI-562-9 or EI-562-10.

3.5.1.3 Weight. The weight of the Receiver-Transmitter shall be not greater than 13.4 pounds.

3.5.1.4 Receiver-Transmitter Assemblies. The Receiver-Transmitter shall contain the following assemblies:

a. Transmitter Module
b. Receiver Module
c. IF Amplifier Module
d. Range Computer
e. Power Supply Module
f. Radio Frequency Filter and Chassis

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3.5.1.4.1 Assembly similarity. The transmitter module, receiver module, IF amplifier module, power supply module and radio frequency filter and chassis assembly for all receiver-transmitters shall be identical in all mechanical and electrical functions. The manufacturer's part identification number of these assemblies shall be identical, except for the unit nameplate data on the radio frequency filter and chassis assembly.

3.5.1.4.2 Range computer. The range computer assembly shall consist of not less than four individual printed circuit card networks. Not more than six printed circuit card networks can be accommodated within the range computer assembly. The variations of Receiver-Transmitter functions shall be accomplished by replacing or adding to the range computer assembly within the unit.

3.5.1.5 Cooling. The Receiver-Transmitter unit shall meet the temperature requirements of this specification without the use of externally supplied forced air.

3.5.1.6 Controls. There shall be no in-flight controls located on the Receiver-Transmitter unit.

3.5.1.7 Electrical connections and signals

*3.5.1.7.1 Electrical connections. Connections to external circuits shall be as shown in EI-562.

3.5.1.7.2 Electrical signal outputs

3.5.1.7.2.1 Altitude signal. The Receiver-Transmitter shall provide a linear analog dc voltage proportional to altitude, having an excursion of 0 to -40 volts, with -40 volts corresponding to 5,000 feet or 1,000 feet and 0 volts corresponding to 0 feet. The signal shall be capable of driving not less than two of the indicators described in 3.5.5, 3.5.6 or 3.5.10 herein. The altitude signal output shall be -46.7 $\pm$ 0.7 volts dc during "no track" conditions.

*3.5.1.7.2.2 Reliability signal. The reliability signal shall consist of a set of relay contacts which are normally closed when the receiver-transmitter is tracking the ground return signal. This set of contacts shall be open during no track conditions. The external loading of the reliability signal shall be not greater than one ampere dc at +28 vdc.

3.5.1.7.2.3 Indicator reference voltage. The Receiver-Transmitter shall provide an indicator reference voltage of -50 $\pm$ 0.5 percent dc. The indicator reference voltage shall be capable of operating two height indicators.

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*3.5.2 Receiver-Transmitter, Radar - RT-804B/APN-171(V)

3.5.2.1 Function. The Receiver-Transmitter shall transmit a radio frequency signal, receive the reflected signal, and derive the proper analog voltage for 0 to 5,000 feet of radar altitude in accordance with the requirements of this specification. The range computer assembly shall consist of four printed circuit card networks.

*3.5.2.1.1 Additional outputs. In addition to the altimeter functions specified in 3.5.1 herein, the RT-804B/APN-171(V) shall provide additional outputs compatible with other aircraft systems. 3.5.2.2 and 3.5.2.3 specify these outputs. The additional circuitry and components which must be added to the RT unit to provide these outputs shall meet and in no way degrade those requirements specified in 3.5.1.

*3.5.2.2 Blanking pulse output. The Receiver-Transmitter shall produce a blanking pulse having the following characteristics:

a.	Load Impedance	1000 ohm typical, 50 ohm min.
b.	Source Impedance	Less than 10 ohms
c.	RF Lead Time	300 $\pm$ 50 ns
d.	Rise Time (10-90%)	20 $\pm$ 15 ns
e.	Fall Time (10-90%)	125 $\pm$ 75 ns
f.	Width (50% points)	900 $\pm$ 350 ns
g.	Amplitude	+10 $\pm$ 2V
h.	Inter-pulse Output	0 $\pm$ 0.5V

*3.5.2.3 External zero altitude adjust. An externally accessible zero altitude adjustment shall be provided on the front of the Receiver-Transmitter to provide altitude adjustment of not less than $\pm$ 5 feet.

3.5.3 Receiver-Transmitter, Radar - RT-805/APN-171(V)

3.5.3.1 Function. The Receiver-Transmitter shall transmit a radio frequency signal, receive the reflected signal, and derive the proper analog voltage for 0 to 1,000 feet of radar altitude in accordance with the requirements of this specification. The range computer assembly shall consist of four printed circuit card networks.

*3.5.4. Receiver-Transmitter, Radar - RT-806A/APN-171(V)

3.5.4.1 Function. The Receiver-Transmitter shall transmit a radio frequency signal, receive the reflected signal, and derive the proper analog voltage for 0 to 1,000 feet of radar altitude in accordance with the requirements of this specification. The range computer assembly shall consist of five printed circuit card networks.

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*3.5.4.1.1 Additional outputs. In addition to the altimeter functions specified in 3.5.1 herein, the RT-806A/APN-171(V) shall provide additional outputs compatible with other aircraft systems. 3.5.4.2 through 3.5.4.6 specify these outputs. The additional circuitry and components which must be added to the Receiver-Transmitter unit to provide these outputs shall meet and in no way degrade those requirements specified in 3.5.1.

*3.5.4.2 Automatic flight control system interface. RT-806A/APN-171(V) shall provide an altitude error output which is referenced to a selected altitude. The error output is used by the Automatic Flight Control System (AFCS) to provide the aircraft with an automatic altitude hold capability.

3.5.4.2.1 Altitude error signal. The altitude error signal is a voltage proportional to the altitude difference between the aircraft altitude as sensed by the altimeter and a reference altitude. The reference altitude is represented by an altitude reference signal as specified in 3.5.4.2.1.8.

3.5.4.2.1.1 Error signal. The altitude error signal shall be a 400 Hz signal with a scale factor of 716 millivolts RMS per foot of error.

3.5.4.2.1.2 Signal null. The altitude error signal shall reach a minimum when the aircraft altitude coincides with the reference altitude. This minimum or null signal shall occur when the altitude signal (see 3.5.1.7.2) is within one foot or 2 percent, whichever is greater, of the reference altitude under standard conditions and one foot or 5 percent, whichever is greater, under service conditions. The null voltage shall be not greater than 716 millivolts RMS and the 400 Hz component shall be not greater than 358 millivolts RMS.

3.5.4.2.1.3 Resolution. The error signal shall have a resolution of 0.5 foot for altitude errors between -50 and +50 feet for reference altitudes extending from 0 to 200 feet.

3.5.4.2.1.4 Accuracy. The slope of the error signal shall be within 5 percent of the 716 millivolt per foot scale factor for all altitude errors between $\pm$ 30 feet.

3.5.4.2.1.5 Linearity. The error signal shall not deviate more than 0.33 feet or 2 percent, whichever is greater, from the best straight line whose slope is within 5 percent of 716 millivolt per foot, for altitude errors between $\pm$ 30 feet.

3.5.4.2.1.6 Phase. The error signal phase angle with reference to the Receiver-Transmitter unit's 400 Hz excitation shall be 0° when the aircraft altitude is greater than the reference altitude and 180° when the aircraft altitude is less than the reference altitude. The phase angle tolerances shall be $\pm$ 7° under standard conditions and $\pm$ 10° under service conditions for altitude errors between $\pm$ 10 feet. For errors greater than $\pm$ 10 feet, the phase angle tolerance shall be $\pm$ 15° under service conditions.

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3.5.4.2.1.7 Load. The altitude error signal shall be capable of operating into a 200,000 ohm resistive load.

3.5.4.2.1.8 Altitude reference signal. The reference altitude, represented by a linear dc voltage proportional to the reference altitude, shall be provided to the Receiver-Transmitter unit. This signal is developed across a 2500 ohm (1 percent resistance tolerance and 2 percent linearity) potentiometer which is excited by a -15 VDC ± 3 percent output from the Receiver-Transmitter unit. The potentiometer is linearly calibrated to 200 feet.

3.5.4.3 Self-test inhibit signal. A 28V (nominal) signal shall be provided to the Receiver-Transmitter unit to inhibit the self-test function of the altimeter. Normal self-test operation shall be restored when this voltage is removed. This signal shall be provided by the AFCS when the radar altimeter coupler is engaged and shall be removed when the coupler is disengaged.

*3.5.4.4 Radar altimeter warning system interface. The RT-806A/APN-171(V) shall provide an altitude proportional signal and an altimeter reliable signal to the radar altimeter warning system (RAWS).

3.5.4.4.1 Proportional signal. A positive, linear dc voltage proportional to altitude shall be available. The range of this signal shall be 1 volt to 11 volts corresponding to 0 feet to 500 feet respectively. The signal shall be not greater than +15 volts dc at altitudes above 500 feet or when the altimeter is not in track.

3.5.4.4.1.1 Accuracy. The proportional signal shall be within 2.5 feet plus 2 percent of actual altitude under standard conditions and within five feet plus 5 percent of actual altitude plus 5 percent of average altitude rate under service conditions in the range of 0 to 500 feet.

3.5.4.4.1.2 Load. The proportional signal shall be as specified when connected to a 10,000 ohm resistive load.

*3.5.4.5 Blanking pulse output. A blanking pulse shall be provided having the following characteristics:

a.	Load Impedance	1000 ohm typical, 50 ohm min.
b.	Source Impedance	Less than 10 ohms
c.	RF Lead Time	300 ± 50 ns
d.	Rise Time (10-90%)	20 ± 15 ns
e.	Fall Time (10-90%)	125 ± 75 ns
f.	Width (50% points)	900 ± 350 ns
g.	Amplitude	+10 ± 2V
h.	Inter-pulse Output	0 ± 0.5V

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*3.5.4.6 External zero altitude adjust. An externally accessible zero altitude adjustment shall be provided on the front of the receiver-transmitter to provide altitude adjustment of not less than ± 5 feet.

3.5.5 Indicator, Height - ID-1345/APN-171(V)

3.5.5.1 Function. The indicator shall indicate altitude from 0 to 5,000 feet by means of a single pointer against a fixed dial. A movable marker positioned against the same dial shall indicate the pre-selected height below which the control signals for the altitude warning device operates. The "no track" condition shall be indicated by a flag coming into view. Altitudes higher than 5,000 feet or the loss of track, shall also be indicated by clockwise movement of the pointer behind the masked portion of the indicator dial. The accuracy, sensitivity and low altitude switch errors shall be within the limits specified in Table I.

3.5.5.2 Form factor. The indicator shall conform to the dimensions of EI-562-12.

3.5.5.3 Weight. The weight of the height indicator shall be not greater than 1.6 pounds.

3.5.5.4 Contents. Each indicator shall include the necessary mechanism to rotate the altitude indicator pointer to a position determined by the system. It shall also include a low altitude limit index.

3.5.5.5 Controls. A control knob located as shown in EI-562-12 shall control the low altitude limit index, actuation of the self-test circuit, and also function as the system power control knob. The low altitude limit index shall be continuously adjustable throughout the range of the equipment. When the equipment is turned off, this index shall be positioned at less than zero feet. When the equipment is turned on, the initial position of the limit index shall be no greater than zero feet. The self-test circuit shall be actuated by depressing the control knob.

*3.5.5.6 Electrical connections. A single multicontact connector shall be mounted on the back of the indicator opposite the dial for all electrical connections. Connections to external circuits shall be as shown on EI-562.

3.5.5.7 Dial markings. The scale markings shall utilize approximately 320 degrees of the dial face. The dial shall be as shown in figure 2. The tip of the pointer shall provide 0.03 inch clearance with the innermost point of the minor graduations. The low altitude limit index shall be in the form of an equilateral triangle and shall be positioned to provide 0.03 inch clearance with the pointer tip.

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3.5.5.8 Indicator protection. Protection shall be provided within the system to avoid damaging the indicator at altitudes above 5000 feet and the system shall resume normal operation while descending through 5000 feet.

3.5.5.9 Low altitude limit. The indicator shall contain the low altitude warning light as specified in 3.4.9.

3.5.5.10 Response. Full scale response of the indicator pointer shall be 2.0 ± 0.3 seconds.

3.5.5.11 Damping. The indicator shall be damped to prevent pointer overshoots greater than one degree. The settling time of the pointer shall be not greater than 0.2 second. Damping shall be 0.6 to 0.85 of critical.

3.5.5.12 Lighting. The indicator shall incorporate integral red lighting in accordance with MIL-L-25467. The indicator lighting shall be compatible with 5 volt and 28 volt ac and dc aircraft lighting systems. For 5 volt lighting, MS24367 lamps with a red coating shall be used. For 28 volt lighting, T-1 size lamps (Los Angeles Miniature Lamp Co. #62 or approved equal) with a red coating shall be used.

3.5.5.12.1 Coverglass and wedges. A reflection reducing film as specified in MIL-C-675 shall be mandatory on the instrument coverglass and optional on intermediate transparent elements between the coverglass and dial.

3.5.5.12.2 Brightness. Lighting shall be in accordance with MIL-L-25467 except that pointer tip brightness shall be greater than that of the nearest exposed indicia.

3.5.5.13 Fogging. The indicator face shall be free of any visible condensation when operating within the environmental conditions specified in 3.3.10. A rapid change in environmental conditions shall not generate condensation on the face of the indicator.

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Table I. ID-1345/APN-171 (V) Indicator characteristics
(0 to 5000 foot unit)

Altitude Range (feet)	Sensitivity (feet)	Pointer Error (maximum)	Low Alt. Errors SW Closing	Switch (feet) SW Opening
0 - 200	± 0.50	+(0.7% of alt) -(1.5ft.+0.1% of alt)	±4	+10
200 -1000	±3	± 10 feet	±20	+47
1000 -5000	± 15	±(32.5 ft.+.88% of alt)	± 107	+267

Table II. ID-1346/APN-171 (V) Indicator characteristics
(0 to 1000 foot unit)

Altitude Range (feet)	Sensitivity (feet)	Pointer Error (maximum)	Low Alt. Errors SW Closing	Switch (feet) SW Opening
0 - 100	±0.30	+(0.7% of alt) -(0.6ft.+0.1% of alt.)	±1	+2.4
100 - 200	±0.50	± 1.5 feet	±2.6	+6.8
200 - 1000	±4.0	±(6.4ft.+0.88% of alt.)	±22.0	+53.4

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3.5.6 Indicator, Height - ID-1346/APN-171(V)

3.5.6.1 Function. The indicator shall be equal to that specified in 3.5.5 herein except that this unit shall indicate altitude from 0 to 1000 feet. The altitude display presentation shall be as shown in Figure 3. The accuracy, sensitivity and low altitude switch errors shall be within the limits specified in Table II.

3.5.7 Antenna - AS-1858/APN-171(V). Two identical passive antennas shall be provided. The antennas shall meet the applicable requirements of the set and the detail requirements specified in 3.5.7.1 through 3.5.7.4.8.

3.5.7.1 Function. The function of the antenna shall be to act as the transducer and impedance matching network between free space and the antenna cables.

*3.5.7.2 Electrical characteristics.

*3.5.7.2.1 Power handling capability. 200 watts peak power maximum at the input connector with .002 duty cycle.

*3.5.7.2.2 Signal rejection. The signal rejection from input to antenna output shall be not less than as follows:

a. 2.5 dB at 8.35 GHz and 8.85 GHz
b. 50 dB from 8.50 GHz to 8.70 GHz

*3.5.7.3 Electrical connections. Connections to external circuits shall be as shown in EI-562.

3.5.7.4 Antenna characteristics.

3.5.7.4.1 Form factor. Each antenna shall be contained within the volume shown on EI-562-13.

3.5.7.4.2 Weight. The weight of each antenna shall be not greater than 1.0 pound.

*3.5.7.4.3 Mounting. Each antenna shall have mounting provisions as shown in EI-562-11. Each antenna shall withstand a differential pressure of one atmosphere.

*3.5.7.4.4 Isolation - The antennas shall provide electrical isolation of not less than 90 dB at a spacing of 3 feet and 85 dB at 2 feet, center-to-center when flush mounted to a ground plane. The ground plane shall consist of a flat sheet of aluminum, 6 feet long on the roll axis and 3 feet wide on the pitch axis with the long sides rolled back from the aperture plane with a 6-inch bend radius. Isolation test to be run at 4300 MHz.

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*3.5.7.4.5 Antenna gain. Each antenna shall provide a minimum E-plane and H-plane gain of 12 dB at 4300 MHz one way. Gain at ±17.5 degrees each side of boresight shall be 10 dB minimum.

*3.5.7.4.6 Electromagnetic fields. Antennas shall be oriented so that the connectors are perpendicular to an imaginary line drawn through the two antenna centers. This arrangement allows the antenna H-planes of radiation to couple and prevents the E-planes of radiation from coupling.

*3.5.7.4.7 Voltage standing wave ratio (VSWR). The free space V.S.W.R. of each antenna and connector shall be not greater than 1.2 at 4300 MHz with a maximum V.S.W.R. of 1.3 at 4250 and 4350 MHz.

*3.5.7.4.8 Side lobes. The side lobe radiation of each antenna shall not exceed the maximum gain envelope described as follows:

a. The E-plane maximum gain shall be defined by a continuous envelope from 50^{0} off boresight to 310^{0} (-50^{0}) off boresight. The maximum gain shall be 25 dB below maximum gain from 90^{0} to 270^{0} (-90^{0}) off boresight, 23.5 dB below maximum gain at $\pm 85^{0}$ off boresight, 22 dB below maximum gain at $\pm 80^{0}$ off boresight, 20 dB below maximum gain at $\pm 70^{0}$ off boresight, and 12 dB below maximum gain at $\pm 50^{0}$ off boresight.

b. The H-plane maximum gain shall be defined by a continuous envelope from 50^{0} off boresight to 310^{0} (-50^{0}) off boresight. The maximum gain shall be 30 dB below maximum gain from 90^{0} to 270^{0} (-90^{0}) off boresight, 26.7 dB below maximum gain $\pm 85^{0}$ off boresight, 25 dB below maximum gain at $\pm 80^{0}$ off boresight, 17 dB below maximum gain at $\pm 60^{0}$ off boresight, 14.8 dB below maximum gain at $\pm 55^{0}$ off boresight, and 13 dB below maximum gain at $\pm 50^{0}$ off boresight.

3.5.8 Mounting MT-4167/APN-171(V)

3.5.8.1 Function. The mounting base, equipped with suitable isolators, shall be provided for the Receiver-Transmitter.

*3.5.8.2 Form factor. The mounting base shall conform to the dimensions in EI-562-11.

3.5.8.3 Weight. Weight of the mounting base shall be not greater than 2.2 pounds.

3.5.8.4 Locking clamps. Locking clamps shall be self locking.

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*3.5.9 Receiver-Transmiter, Radar RT-829A/APN-171(V)

3.5.9.1 Function. The Receiver-Transmitter shall transmit a radio frequency signal, receive the reflected signal, and derive the proper analog voltage for 0 to 5,000 feet of radar altitude in accordance with the requirements of this specification. The range computer assembly shall consist of six printed circuit card networks.

*3.5.9.1.1 Additional outputs. In addition to the altimeter functions specified in 3.5.1 herein, the RT-829A/APN-171(V) shall provide additional outputs compatible with other aircraft systems. 3.5.9.2 through 3.5.9.6 specify these outputs. The additional circuitry and components which shall be added to the RT unit to provide these outputs shall meet and in no way degrade those requirements specified in 3.5.1.

*3.5.9.2 Automatic flight control system interface. The RT-829A/APN-171(V) shall provide an altitude error output referenced to a selected altitude, and an altitude rate output. These outputs are used by the Automatic Flight Control System (AFCS) to provide the aircraft with an automatic altitude hold capability.

3.5.9.2.1 Altitude error signal. The altitude error signal is a voltage proportional to the altitude difference between the aircraft altitude as sensed by the altimeter and a reference altitude. The reference altitude is represented by an altitude reference signal as specified in 3.5.9.2.1.8.

3.5.9.2.1.1 Error signal. The altitude error signal shall be a 400 Hz signal with a scale factor of 70 millivolts RMS per foot of error.

3.5.9.2.1.2 Signal null. The altitude error signal shall reach a minimum when the aircraft altitude coincides with the reference altitude. This minimum or null signal shall occur when the altitude signal (see 3.5.1.7.2.1) is within one foot or 2 percent, whichever is greater, of the reference altitude under standard conditions and one foot or 5 percent, whichever is greater, under service conditions. The null voltage shall not exceed 70 millivolts RMS and the 400 Hz component shall be not greater than 35 millivolts RMS.

3.5.9.2.1.3 Resolution. The error signal shall have a resolution of 0.5 foot for altitude errors between -50 and +50 feet for reference altitude extending from 0 to 500 feet.

3.5.9.2.1.4 Accuracy. The slope error of the error signal shall be within 5 percent of the 70 millivolt per foot scale factor for all altitude errors between ± 30 feet.

3.5.9.2.1.5 Linearity. The error signal shall not deviate more than 0.33 feet or 2 percent, whichever is greater, from the best straight line whose slope is within 5 percent of 70 millivolt per foot, for altitude errors between ± 30 feet.

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3.5.9.2.1.6 Phase. The error signal phase angle with reference to the Receiver-Transmitter unit's 400 Hz excitation shall be 0° when the aircraft altitude is less than the reference altitude and 180° when the aircraft altitude is greater than the reference altitude. The phase angle tolerance shall be ± 7° under standard conditions and ± 10° under service conditions for altitude errors between ± 10 feet. For errors greater than ± 10 ft the phase angle tolerance shall be ± 15° under service conditions.

3.5.9.2.1.7 Load. The altitude error signal shall be capable of operating into a 20,000 ohm resistive load.

3.5.9.2.1.8 Altitude reference signal. The reference altitude, represented by a linear dc voltage proportional to the reference altitude, shall be provided to the Receiver-Transmitter unit. This signal is developed across a 2500 ohm (1 percent resistance tolerance and 1 percent linearity) potentiometer which is excited by a -15 VDC ±3 percent output from the RT unit. This potentiometer is linearly calibrated to 500 feet.

3.5.9.2.2 Altitude rate signal. The altitude rate signal shall be a voltage proportional to the vertical velocity of the aircraft relative to the ground.

3.5.9.2.2.1 Rate signal. The altitude rate signal shall be a 400 Hz voltage with a scale factor of 150 millivolt RMS per foot per second.

3.5.9.2.2.2 Signal null. The altitude rate signal shall reach a minimum within ± 0.75 ft/sec under standard conditions and ± 1.25 ft/sec under service conditions of zero aircraft altitude rate. This minimum shall be not greater than 120 mv RMS and shall have a 400 Hz component that is not greater than 50 mv RMS.

3.5.9.2.2.3 Resolution. The rate signal shall have a resolution of 0.5 ft/sec for altitude rates between ± 10 ft/sec.

3.5.9.2.2.4 Accuracy. The slope error of the rate signal shall be within 5 percent of the 150 millivolt/ft/sec scale factor for rates up to ± 10 ft/sec.

3.5.9.2.2.5 Linearity. The rate signal shall not deviate more than 0.33 ft/sec from the best straight line whose slope is within 5 percent of 150 mv/ft/sec, for altitude rates between ± 10 ft/sec.

3.5.9.2.2.6 Phase. The rate signal phase, referenced to the 400 Hz excitation to the Receiver-Transmitter unit, shall be of 0° for a descending rate and 180° for an ascending rate. The phase angle tolerances shall be ± 7° under standard conditions and ± 10° under service conditions for altitude rates between ± 10° ft/sec. For rates greater than ± 10 ft/sec, the phase angle tolerance shall be ± 15° under standard conditions.

3.5.9.2.2.7 Loading. The altitude rate signal shall operate into a 10,000 ohm resistive load.

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3.5.9.3 Self-test inhibit signal. A 28V (nominal) signal shall be provided to the Receiver-Transmitter unit to inhibit the self-test function of the altimeter. Normal self-test operation shall be restored when this voltage is removed. This signal shall be provided by the AFCS when the radar altimeter coupler is engaged and shall be removed when the coupler is disengaged.

*3.5.9.4 Landing gear warning system interface. The RT-829A/APN-171(V) shall provide an altimeter ON and a -50 V signal for the aircraft's landing gear warning system.

3.5.9.4.1 Power ON signal. The Receiver-Transmitter unit shall provide a +28 (+5, -8) VDC output signal when the altimeter power is ON and 0 ± 1 VDC when altimeter power is OFF. This signal shall be capable of operating a 1000 ohm load.

3.5.9.4.2 -50 V signal. The Receiver-Transmitter unit shall provide a -50 VDC output through a 50,000 ohm nominal resistance.

*3.5.9.5 Blanking pulse output. A blanking pulse shall be provided having the following characteristics:

a.	Load Impedance	1000 ohm typical, 50 ohm min.
b.	Source Impedance	Less than 10 ohms
c.	RF Lead Time	300 ± 50 ns
d.	Rise Time (10-90%)	20 ± 15 ns
e.	Fall Time (10-90%)	125 ± 75 ns
f.	Width (50% points)	900 ± 350 ns
g.	Amplitude	+10 ± 2V
h.	Inter-pulse Output	0 ± 0.5V

*3.5.9.6 External zero altitude adjust. An externally accessible zero altitude adjustment shall be provided on the front of the receiver-transmitter to provide altitude adjustment of not less than ± 5 feet.

3.5.10 Indicator Height ID-1725/APN-171(V)

3.5.10.1 Function. The indicator shall be equal to that specified in 3.5.5 herein except that the lighting color shall be white in accordance with MIL-L-27160.

3.5.11 Receiver-Transmitter, Radar RT-1006/APN-171(V)

3.5.11.1 Function. The Receiver-Transmitter shall transmit a radio frequency signal, receive the reflected signal, and derive the proper analog voltage for 0 to 5,000 feet of radar altitude in accordance with the requirements of this specification. The range computer assembly shall consist of six printed circuit card networks.

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3.5.11.1.1 Additional outputs. In addition to the altimeter functions specified in 3.5.1 herein, the RT-1006/APN-171(V) shall provide additional outputs compatible with other aircraft systems. 3.5.11.2 through 3.5.11.4.2 specify these outputs. The additional circuitry and components which must be added to the Receiver-Transmitter unit to provide these outputs shall meet and in no way degrade those requirements specified in 3.5.1.

3.5.11.2 Automatic flight control system interface. The RT-1006/APN-171(V) shall provide an altitude error output referenced to a selected altitude, and an altitude rate output. These outputs are used by the Automatic Flight Control System (AFCS) to provide the aircraft with an automatic altitude hold capability.

3.5.11.2.1 Altitude error signal. The altitude error signal is a voltage proportional to the altitude difference between the aircraft altitude as sensed by the altimeter and a reference altitude. The reference altitude is represented by an altitude reference signal as specified in 3.5.11.2.1.9.

3.5.11.2.1.1 Error signal. The altitude error signal shall be a 400 Hz square wave signal with a scale factor of 70 millivolts RMS per foot of error.

3.5.11.2.1.2 Signal null. The altitude error signal shall reach a minimum when the aircraft altitude coincides with the reference altitude. This minimum or null signal shall occur when the altitude signal (see 3.5.1.7.2.1) is within one foot or 2 percent, whichever is greater, under service conditions. The null voltage shall be not greater than 70 millivolts RMS and the 400 Hz component shall be not greater than 35 millivolts RMS.

3.5.11.2.1.3 Resolution. The error signal shall have a resolution of 0.5 foot for altitude errors between -50 and +50 feet for reference altitude extending from 0 to 500 feet.

3.5.11.2.1.4 Saturation. Saturation shall not occur within ±50 feet (minimum) of the selected altitude. For altitude errors of greater than ±50 feet, it shall be permissible for the differential gain of the error circuit to be zero (i.e., saturation is permissible). However, the saturated waveform shall remain square, and no phase shift shall be introduced as a result of saturation.

3.5.11.2.1.5 Accuracy. The slope error of the error signal shall be within 5 percent of the 70 millivolts per foot scale factor for all altitude errors between ±30 feet.

3.5.11.2.1.6 Linearity. The error signal shall not deviate more than 0.33 feet or 2 percent, whichever is greater, from the best straight line whose slope is within 5 percent of 70 millivolt per foot, for altitude errors between ± 30 feet.

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3.5.11.2.1.7 Phase. The error signal phase angle with reference to the Receiver-Transmitter unit's 400 Hz excitation shall be 0° when the aircraft altitude is less than the reference altitude and 180° when the aircraft altitude is greater than the reference altitude. The phase angle tolerance between the Receiver-Transmitter unit's 400 Hz reference and the error signal shall be ±3° under all conditions.

3.5.11.2.1.8 Load. The altitude error signal shall be capable of operating into a 2000 ohm resistive load or greater.

3.5.11.2.1.9 Altitude reference signal. The reference altitude, represented by a linear dc voltage proportional to the reference altitude, shall be provided to the Receiver-Transmitter unit. This signal is developed across a 2500 ohm (1 percent resistance tolerance and 1 percent linearity) potentiometer which is excited by a -15VDC ± 3 percent output from the Receiver-Transmitter unit. This potentiometer is linearly calibrated to 500 feet.

3.5.11.2.2 Altitude rate signal. The altitude rate signal shall be a voltage proportional to the vertical velocity of the aircraft relative to the ground.

3.5.11.2.2.1 Rate signal. The altitude rate signal shall be a 400 Hz voltage with a scale factor of 150 millivolt RMS per foot per second.

3.5.11.2.2.2 Signal null. The altitude rate signal shall reach a minimum within ± 0.75 ft/sec under standard conditions and ±1.25 ft/sec under service conditions of zero aircraft altitude rate. This minimum shall be not greater than 120 mv RMS and shall have a 400 Hz component that is not greater than 50 mv RMS.

3.5.11.2.2.3 Resolution. The rate signal shall have a resolution of 0.5 ft/sec for altitude rates between ± 10 ft/sec.

3.5.11.2.2.4 Accuracy. The slope error of the rate signal shall be within 5 percent of the 150 millivolt/ft/sec scale factor for rates up to ± 10 ft/sec.

3.5.11.2.2.5 Linearity. The rate signal shall not deviate more than 0.33 ft/sec from the best straight line whose slope is within 5 percent of 150 mv/ft/sec, for altitude rates between ± 10 ft/sec.

3.5.11.2.2.6 Phase. The rate signal phase, referenced to the 400 Hz excitation to the Receiver-Transmitter unit, shall be of 0° for a descending rate and 180° for an ascending rate. The phase angle tolerances shall be ±7° under standard conditions and ± 10° under service conditions for altitude rates between ± 10 ft/sec. For rates greater than ± 10 ft/sec, the phase angle tolerance shall be ± 15° under standard conditions.

3.5.11.2.2.7 Loading. The altitude rate signal shall operate into a 10,000 ohm resistive load.

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3.5.11.3 Self-test inhibit signal. A 28V (nominal) signal shall be provided to the Receiver-Transmitter unit to inhibit the self-test function of the altimeter. Normal self-test operation shall be restored when this voltage is removed. This signal shall be provided by the AFCS when the radar altimeter coupler is engaged and shall be removed when the coupler is disengaged.

3.5.11.4 Landing gear warning system interface. The RT-1006/APN-171(V) shall provide an altimeter ON and a -50 V signal for the aircraft's landing gear warning system.

3.5.11.4.1 Power ON signal. The Receiver-Transmitter unit shall provide a +28 (+5, -8) VDC output signal when the altimeter power is ON and 0 ± 1 VDC when altimeter power is OFF. This signal shall be capable of operating a 1000 ohm load.

3.5.11.4.2 -50 V signal. The Receiver-Transmitter unit shall provide a -50 VDC output through a 50,000 ohm nominal resistance.

3.5.12 Receiver-Transmitter, Radar RT-1062A/APN-171(V)

3.5.12.1 Function. The Receiver-Transmitter shall transmit a radio frequency signal, receive the reflected signal, and derive the proper analog voltage for 0 to 5,000 feet of radar altitude in accordance with the requirements of this specification. The range computer assembly shall consist of six printed circuit card networks.

3.5.12.1.1 Additional outputs

3.5.12.1.1.1 External range rate selection. In addition to the range rate tracking capability in 3.4.4, circuitry shall be provided for an externally selectable nominal 50-foot per second range rate tracking capability to provide smooth low speed operation over water.

*3.5.12.1.1.2 Additional capabilities. In addition to the altimeter functions specified in 3.5.1, the RT-1062A/APN-171(V) shall provide additional outputs compatible with other aircraft systems. 3.5.12.2 through 3.5.12.7 specify these outputs. The additional circuitry and components which must be added to the RT unit to provide these outputs shall meet and in no way degrade those requirements specified in 3.5.1.

*3.5.12.2 Flight control system interface. The RT-1062A/APN-171(V) shall provide an altitude error output referenced to a selected altitude, and an altitude rate output. These outputs are used by the Automatic Flight Control System (AFCS) to provide the aircraft with an automatic altitude hold capability.

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3.5.12.2.1 Altitude error signal. The altitude error signal is a voltage proportional to the altitude difference between the aircraft altitude as sensed by the altimeter and a reference altitude. The reference altitude is represented by an altitude reference signal as specified in 3.5.12.2.1.9.

3.5.12.2.1.1 Error signal. The altitude error signal shall be a 400 Hz square wave signal with a scale factor of 70 millivolts RMS per foot of error.

3.5.12.2.1.2 Signal null. The altitude error signal shall reach a minimum when the aircraft altitude coincides with the reference altitude. This minimum or null signal shall occur when the altitude signal (see 3.5.1.7.2.1) is within one foot or 2 percent, whichever is greater, of the reference altitude under standard conditions and one foot or 5 percent, whichever is greater, under service conditions. The null voltage shall be not greater than 70 millivolts RMS and the 400 Hz component shall be not greater than 35 millivolts RMS.

3.5.12.2.1.3 Resolution. The error signal shall have a resolution of 0.5 foot for altitude errors between -50 and +50 feet for reference altitude extending from 0 to 500 feet.

3.5.12.2.1.4 Saturation. Saturation shall not occur within ± 50 feet (minimum) of the selected altitude. For altitude errors of greater than ± 50 feet, it shall be permissible for the differential gain of the error circuit to be zero (i.e., saturation is permissible). However, the saturated waveform shall remain square, and no phase shift shall be introduced as a result of saturation.

3.5.12.2.1.5 Accuracy. The slope error of the error signal shall be within 5 percent of the 70 millivolt per foot scale factor for all altitude errors between ± 30 feet.

3.5.12.2.1.6 Linearity. The error signal shall not deviate more than 0.33 feet or 2 percent, whichever is greater, from the best straight line whose slope is within 5 percent of 70 millivolt per foot, for altitude errors between ± 30 feet.

3.5.12.2.1.7 Phase. The error signal phase angle with reference to the Receiver-Transmitter unit's 400 Hz excitation shall be 0° when the aircraft altitude is less than the reference altitude and 180° when the aircraft altitude is greater than the reference altitude. The phase angle tolerance between the Receiver-Transmitter unit's 400 Hz reference and the error signal shall be $\pm 3^{o}$ under all conditions.

3.5.12.2.1.8 Load. The altitude error signal shall be capable of operating into a 2,000 ohm resistive load or greater.

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3.5.12.2.1.9 Altitude reference signal. The reference altitude, represented by a linear dc voltage proportional to the reference altitude, shall be provided to the Receiver-Transmitter unit. This signal is developed across a 2500 ohm (1 percent resistance tolerance and 1 percent linearity) potentiometer which is excited by a -15 VDC ± 3 percent output from the Receiver-Transmitter unit. This potentiometer is linearly calibrated to 500 feet.

3.5.12.2.2 Altitude rate signal. The altitude rate signal shall be a voltage proportional to the vertical velocity of the aircraft relative to the ground.

3.5.12.2.2.1 Rate signal. The altitude rate signal shall be a 400 Hz voltage with a scale factor of 150 millivolt RMS per foot per second.

3.5.12.2.2.2 Signal null. The altitude rate signal shall reach a minimum within ± 0.75 ft/sec under standard conditions and ±1.25 ft/sec under service conditions of zero aircraft altitude rate. This minimum shall be not greater than 120 mv RMS and shall have a 400 Hz component that is not greater than 50 mv RMS.

3.5.12.2.2.3 Resolution. The rate signal shall have a resolution of 0.5 ft/sec for altitude rates between ± 10 ft/sec.

3.5.12.2.2.4 Accuracy. The slope error of the rate signal shall be within 5 percent of the 150 millivolt/ft/sec scale factor for rates up to ± 10 ft/sec.

3.5.12.2.2.5 Linearity. The rate signal shall not deviate more than 0.33 ft/sec from the best straight line whose slope is within 5 percent of 150 mv/ft/sec, for altitude rates between ± 10 ft/sec.

3.5.12.2.2.6 Phase. The rate signal phase, referenced to the 400 Hz excitation to the Receiver-Transmitter unit, shall be of 0° for a descending rate and 180° for an ascending rate. The phase angle tolerances shall be ± 7° under standard conditions and ± 10° under service conditions for altitude rates between ± 10 ft/sec. For rates greater than ± 10 ft/sec, the phase angle tolerance shall be ± 15° under standard conditions.

3.5.12.2.2.7 Loading. The altitude rate signal shall operate into a 10,000 ohm resistive load.

3.5.12.3 Self-test inhibit signal. A 28V (nominal) signal shall be provided to the Receiver-Transmitter unit to inhibit the self-test function of the altimeter. Normal self-test operation shall be restored when this voltage is removed. This signal shall be provided by the AFCS when the radar altimeter coupler is engaged and shall be removed when the coupler is disengaged.

*3.5.12.4 Landing gear warning system interface. The RT-1062A/APN-171(V) shall provide an altimeter ON and a -50 V signal for the aircraft's landing gear warning system.

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3.5.12.4.1 Power ON signal. The Receiver-Transmitter unit shall provide a +28 (+5, -8) VDC output signal when the altimeter power is ON and 0 + 1 VDC when altimeter power is OFF. This signal shall be capable of operating a 1000 ohm load.

3.5.12.4.2 -50 V signal. The Receiver-Transmitter unit shall provide a -50 VDC output through a 50,000 ohm nominal resistance.

*3.5.12.5 Aircraft RAD ALT (LAND/SEA) switch. The receiver-transmitter shall interface with an aircraft switch which, when placed in the SEA mode, shall supply a zero (0) ohm ground signal to the receiver-transmitter to lock the receiver-transmitter to a 50 ± 15 ft/sec tracking rate.

*3.5.12.6 Blanking pulse output. A blanking pulse shall be provided having the following characteristics:

a.	Load Impedance	1000 ohm typical, 50 ohm min.
b.	Source Impedance	Less than 10 ohms
c.	RF Lead Time	300 ± 50 ns
d.	Rise Time (10-90%)	20 ± 15 ns
e.	Fall Time (10-90%)	125 ± 75 ns
f.	Width (50% points)	900 ± 350 ns
g.	Amplitude	+10 ± 2V
h.	Inter-pulse Output	0 ± 0.5V

*3.5.12.7 External zero altitude adjust. An externally accessible zero altitude adjustment shall be provided on the front of the receiver-transmitter to provide altitude adjustment of of not less than ± 5 feet.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.2)
b. Initial Production Tests (see 4.3)
c. Acceptance Tests (see 4.4)
d. Life Tests (see 4.5)

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4.2 First article inspection. First article tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. First article tests shall be accomplished under the approved test procedure of 4.6.

4.2.1 First article test data. The contractor shall make available all data collected in conducting these tests to the procuring activity for review.

4.2.2 Scope of tests. First article tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. First article tests shall include environmental tests in accordance with the procedures of MIL-T-5422 and interference tests in accordance with MIL-I-6181.

4.2.3 First article approval. Approval of the first article sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the first article sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The first article sample shall not be considered as one of the equipment under the contract.

4.2.4 Production equipment. Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equal to the approved first article sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial production tests. One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests (see 6.2.1). This equipment will be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on the equipment prior to starting the Initial Production Tests. The first article sample shall not be selected for this test.

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4.3.1 Scope of tests. The equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equal to the previously approved first article sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements of this specification and the contract.

4.3.2 Accessory material. In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and include data necessary to test the equipment.

4.3.3 Initial production sample approval. Approval of the Initial Production Sample will be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.4 Acceptance Tests. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Accpeptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance test shall consist of the following:

a. Individual Tests
b. Sampling Tests
c. Reliability Assurance Tests
d. Special Tests

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4.4.1 Individual tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of Product
b. Operational Test
c. Manufacturing Run in Test

4.4.1.1 Examination of product. Each equipment shall be examined carefully to determine that the material and workmanship requirements of this specification and the contract have been met.

4.4.1.2 Operational test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure specified equipment operation.

4.4.1.3 Manufacturing run in test. Each equipment shall be operated under the conditions specified herein for a period of 10 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least 10 hours of operation.

a.	Temperature	Ambient room
b.	Humidity	Ambient room
c.	Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of $\pm$ 3 g's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 10 hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of specified performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the 10 hour period of operation following the 10 minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10 minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be

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kept of all failures. The 10 hour period specified above may be composed of two 5 hour periods to conform with standard working hours.

4.4.2 Sampling tests. Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0#
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

Sampling tests are not required when Reliability Assurance Tests are conducted.

#NOTE: If by contract action the initial Production Test is deleted, a Sample Test shall also be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of tests. As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

c. Manufacturing run in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability assurance tests. Reliability assurance tests shall be conducted using MIL-STD-781. Tests as required by both the qualification phase and the production acceptance (Sampling) Phase shall be conducted. Classification of failures shall be in accordance with MIL-STD-781 and MIL-STD-2074.

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4.4.3.1 Qualification phase. Prior to the acceptance of equipments under the contract or order, not less than three equipments shall be tested as specified in MIL-STD-781 under the section entitled "Qualification Phase of Production Reliability Tests". The number of equipments to be used shall be not greater than those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan III shall be used.

4.4.3.2 Reliability production acceptance phase tests. Samples of the production equipment shall be tested as specified in MIL-STD-781 (as modified herein), under the Section entitled "Sampling Phase of Production Reliability Tests". For the Sampling Phase (or All-Equipment Test), Test Level E shall be used.

4.4.3.2.1 All equipment test. Each equipment produced shall be tested for 50 hours. Prior to the 50 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures thereon (not counting burn-in failures of burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be made available to the procuring activity. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.3.3 Test details. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be made available to and accepted by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 Special tests. Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

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4.4.4.1 Special test schedule. Selection of equipments for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of tests. Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and make it available to the procuring activity for acceptance prior to conducting the tests.

4.4.5 Equipment failure. Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or design defect.

c. Make available to the procuring activity for approval, proposed corrective action intended to reduce the possiblity of the same failure(s) occurring in future tests.

d. Where practical, include a test in the individual test to check all equipment for this requirement until assurance is obtained that the defect has been corrected.

4.5 Life test. The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following: (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test conditions. The life test shall be conducted under the following simulated service conditions:

a.	Temperature	Normal room
b.	Altitude	Normal ground (0-5000 ft.)
c.	Humidity	Room ambient
d.	A.C. Voltage	115 ± 5 volts (at lowest applicable frequency)
e.	D.C. Voltage	27.5 ± 2.0 volts

4.5.2 Test periods. The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance check. At approximately 8 hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to acceptance by the procuring activity.

4.5.4 Test data. The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies of failures. In the event of part failure, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

a. Part type number
b. The circuit reference symbol number
c. The part function
d. Name of the manufacturer
e. Nature of the failure
f. The number of hours which the part operated prior to failure.

4.5.4.1 Failure report. In the event of a failure, the government inspector shall be notified immediately. A report shall be made available to the procuring activity upon completion of test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

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4.6 Test procedures. The procedures used for conducting first article tests, acceptance tests and life tests shall be prepared by the contractor and made available to the procuring activity for review and acceptance. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Presubmission testing. No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements of this specification and the contract.

5. PREPARATION FOR DELIVERY

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in appendix D of MIL-STD-794.

6. NOTES

6.1 Intended use. The Altimeter Set is intended to indicate the altitude of the carrying aircraft above the surface of the earth. The Altimeter Set should be compatible with numerous types of aircraft systems. The alternate receiver-transmitters and height indicators are provided herein to implement this versatility while minimizing the logistic support required. Note -- The versatility of the receiver-transmitter is provided by providing various range computer assemblies which can be used to perform necessary interface functions with other aircraft systems. The basic altimeter function of indicating absolute altitude can be accommodated within the range computer while still providing growth potential within the unit. This growth capability can be used to meet future system requirements by adding one or two printed circuit card assemblies to the range computer assembly without any other modifications to the receiver-transmitter. The logistic support is minimized because all receiver-transmitter assemblies other than the range computers are identical and interchangeable.

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6.2 Ordering data. Acquisition documents should specify the following:

6.2.1 Acquisition requirements.

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (See 5.1.).

6.2.2 Data requirements. When this specification is used in acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.
3.3.2.2	Plan, Maintenance	UDI-L-21013C
4.2.1	Report, First Article (Preproduction) Test	UDI-T-21349
4.3.3, 4.4.5, 4.5.4.1	Report, Failure	UDI-R-21141A
4.4	Reports, Test	DI-T-2072
4.4.1.3, 4.5.4	Records, Inspection and Test	UDI-T-22706A
4.4.3.2.1	Progress Report	DI-A-21457A
4.4.4.2, 4.5.3, 4.6	Procedures, Test	UDI-T-21347

(Data item description related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence should apply:

(a) Contract - The contract should have precedence over any specification.

(b) This Specification - This specification should have precedence over all applicable subsidiary specifications. Any deviation from subsidiary specifications where applicable should be specifically approved in writing by the procuring activity.

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(c) Referenced Specifications - Any referenced specification should have precedence over all applicable subsidiary specifications referenced herein. All referenced specifications should apply to the extent specified.

6.4 Performance objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirement of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from use of materials, parts and processes other than those specified in specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Non-repairable subassemblies. As a general rule, non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-Between-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.6 Type designations. The parentheses ( ) when used in the equipment type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Revisions. In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

*6.8 Associated equipment. Radar Altimeter Warning Set AN/APQ-107. SH-3D Automatic Stabilization Equipment. CH-53 Automatic Flight Control System. HH53B/LNRS and HH53C/NRS Automatic Flight Control System. RH-53D Automatic Flight Control System, Landing Gear Warning System and Rad Alt (LAND/SEA) Switch.

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6.9 First article. When a first article inspection is required, the item will be tested and should be (a first article sample), (a first production item), (a sample selected from the first 10 production items) (or it may be a standard production item from the contractor's current inventory as specified in 4.1.2). The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the documents first article.

Preparing activity
Navy (AS)
(Project 5826-N187)

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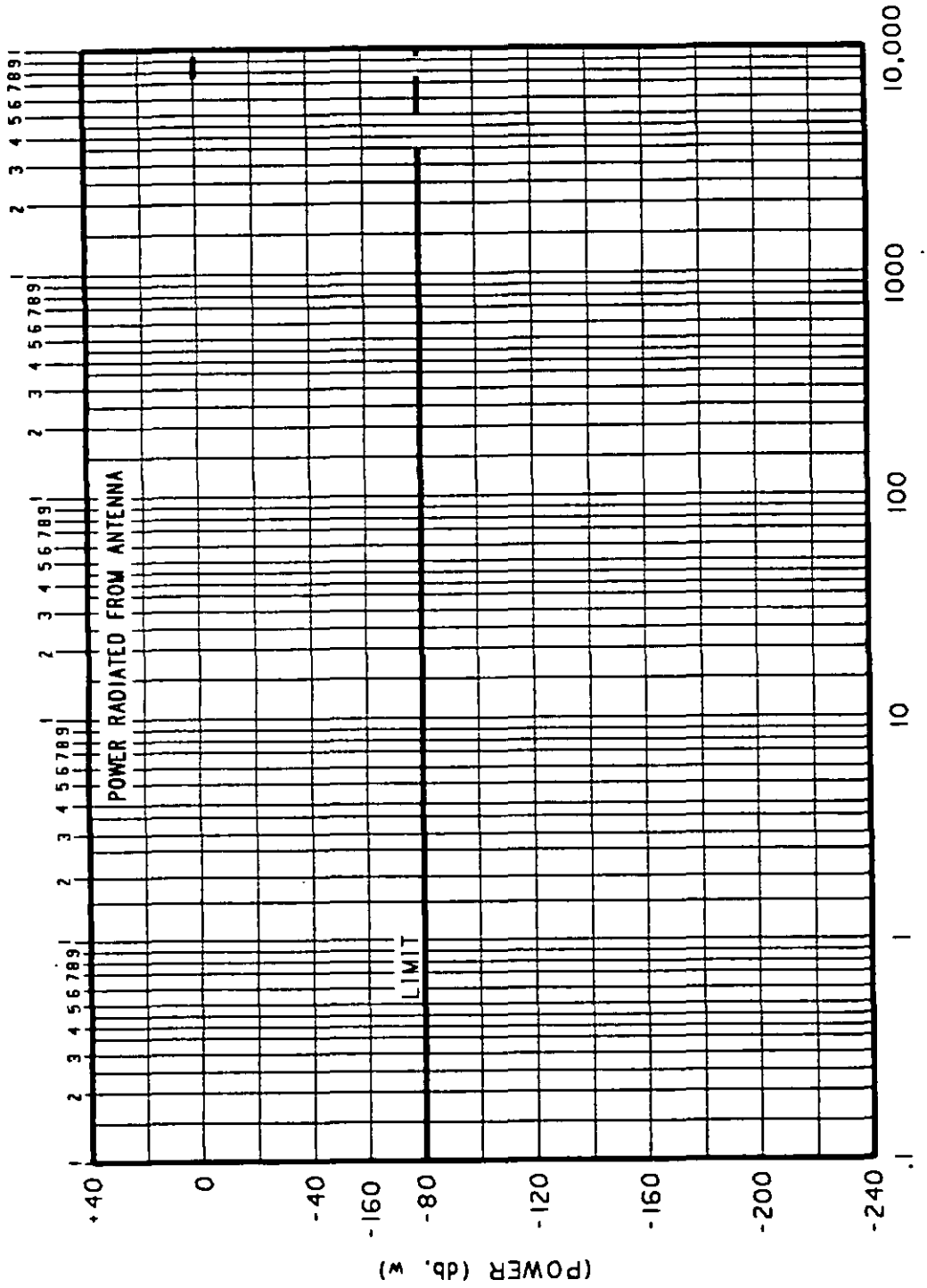
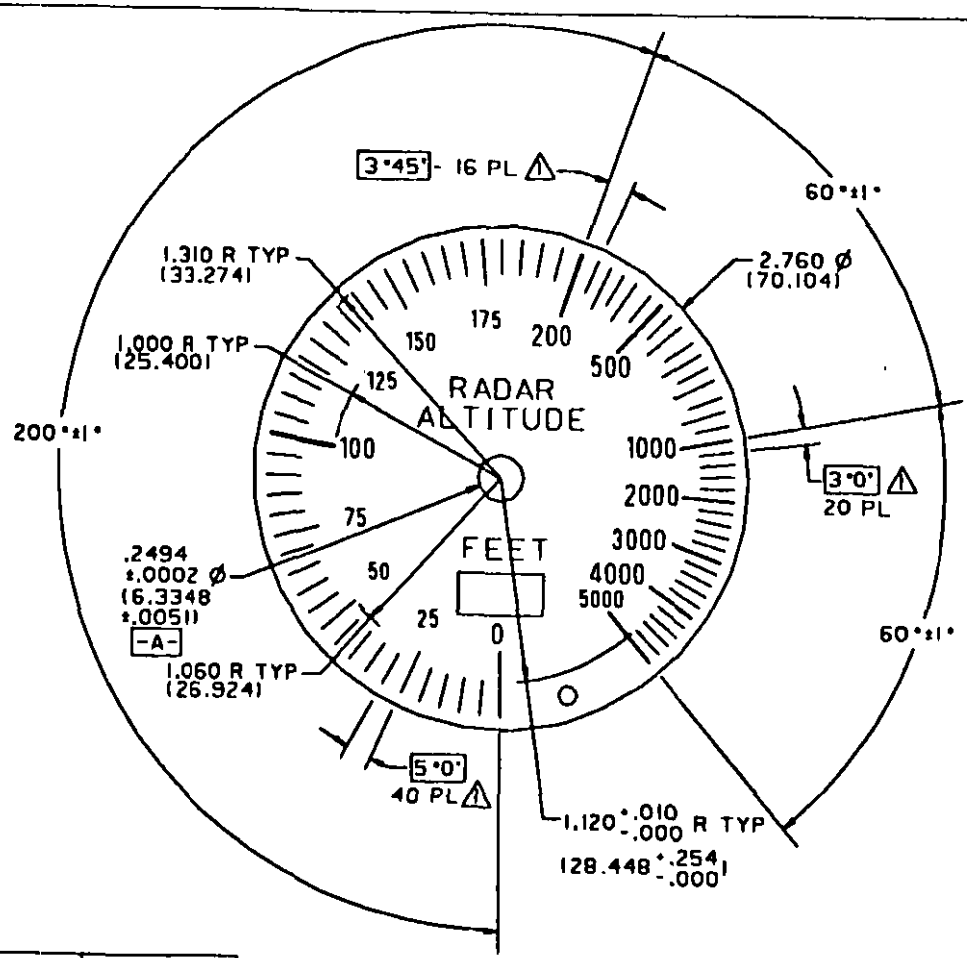


FIGURE 1. POWER RADIATED FROM ANTENNA

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MARKINGS	HEIGHT ±.005 (.127)	WIDTH ±.005 (.127) 2
RADAR ALTITUDE	.125 (3.175)	
FEET & 3	.080 (2.032)	
NUMERALS EXCEPT 3	.140 (3.556)	
MAJOR GRAD		.030 (.762)
MINOR GRAD		.025 (.635)
SUB-MINOR GRAD		.020 (.508)

NOTES:

1 - ALL INDICES TO BE ⌖ Ⓐ Ⓢ .010 TOTAL

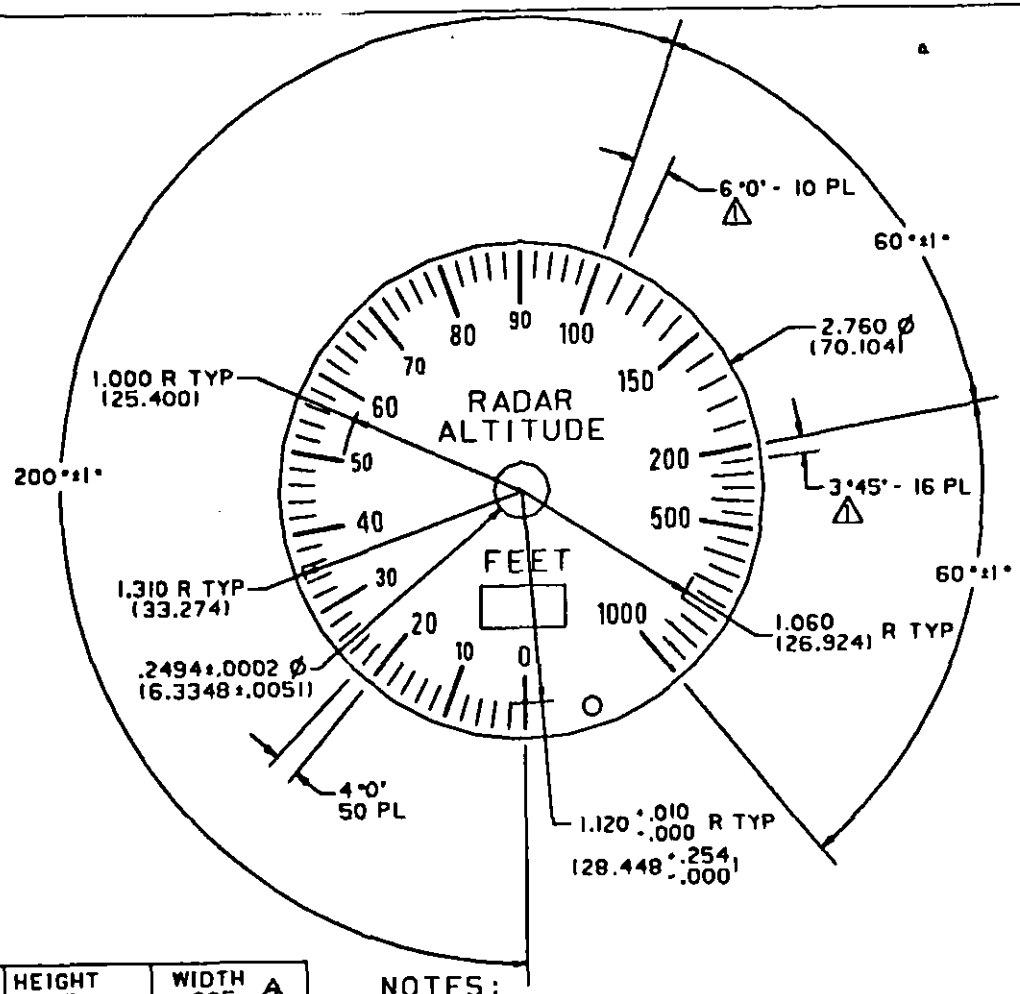
2 - THIS IS AN OVERALL DIM TOL ANY ONE INDEX CANNOT VARY IN WIDTH BY MORE THAN 1/2 OF THIS TOL. AVG LINE WIDTH OF ADJACENT INDICES HAVING THE SAME DIM MUST NOT VARY BY MORE THAN 1/2 OF THIS TOL.

3 - FOR 25, 50, 75, 125, 150, & 5000 NUMERALS

4 - ALL DIMENSION TOLERANCES ARE ±.005 (.127) UNLESS OTHERWISE NOTED. PARENTHETICAL DIMENSIONS ARE IN MILLIMETERS

FIGURE 2. INDICATOR HEIGHT - DIAL MARKINGS ID-1345/APN-171(V) AND ID-1725/APN-171(V)

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MARKINGS	HEIGHT ±.005 (.127)	WIDTH ±.005 (.127) ⚠2
RADAR ALTITUDE	.125 (3.175)	
FEET & 10, 30, 50, 70, & 90	.080 (2.032)	
NUMERALS 0, 20, 40, 60, 80, 150, 200, 500, 1000, & 100	.140 (3.556)	
MAJOR GRAD		.030 (.762)
MINOR GRAD		.025 (.635)
SUB-MINOR GRAD		.020 (.508)

NOTES:

⚠1 - THE CENTERS OF ALL INDICES AT 1.130 R MUST BE ON TRUE POSITION WITHIN .005 (.127) AS MEAS ON AN ARC HAVING THE SAME CENTER AS THE .2494 REF HOLE. TOL FOR CENTER LINE OF REMAINDER OF EACH INDEX IS ±0°15'

⚠2 - THIS TOL IS AN OVERALL DIMENSION TOL. ANY ONE INDEX CANNOT VARY IN WIDTH BY MORE THAN 1/2 OF THIS TOL. AVERAGE LINE WIDTH OF ADJACENT INDICES HAVING THE SAME DIMENSION MUST NOT VARY BY MORE THAN 1/2 OF THIS TOLERANCE

3 - ALL DIMENSION TOLERANCES ARE ±.005 (.127) UNLESS OTHERWISE NOTED. PARENTHETICAL DIMENSIONS ARE IN MILLIMETERS

FIGURE 3. INDICATOR HEIGHT - DIAL MARKINGS ID-1346/APN-171(V)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-81296D (AS)	Altimeter Set, Electronic, AN/APN-171(V)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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